

Topological Sort

Solved

Difficulty: Medium Accuracy: 56.52% Submissions: 362K+ Points: 4 Average Time: 15m

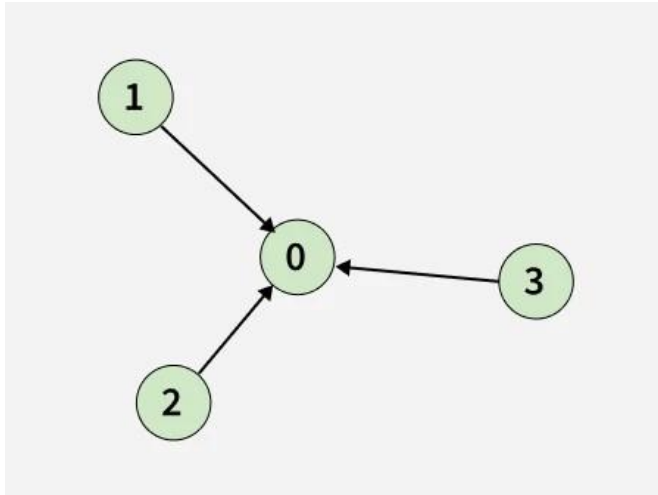
Given a Directed Acyclic Graph (DAG) of V (0 to $V-1$) vertices and E edges represented as a 2D list of `edges[i] = [u, v]` denotes a directed edge $u \rightarrow v$. Return the topological sort for the given graph.

Topological sorting for Directed Acyclic Graph (DAG) is a linear ordering of vertices such that for every directed edge $u \rightarrow v$, vertex u comes before v in the ordering.

Note: As there are multiple Topological orders possible, you may return any of them. If your returned Topological sort is correct then the output will be true else false.

Examples:

Input: $V = 4$, $E = 3$, `edges[i] = [[3, 0], [1, 0], [2, 0]]`



Output: true

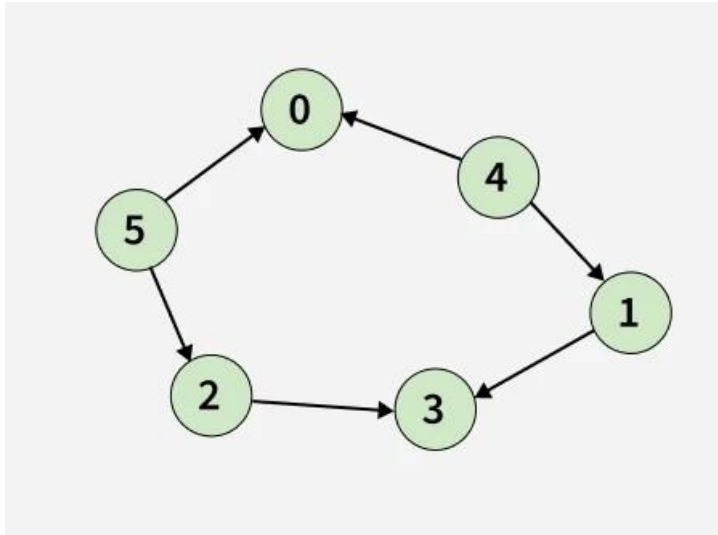
Explanation: The output true denotes that the order is valid. Few valid Topological orders for the given graph are:

[3, 2, 1, 0]

[1, 2, 3, 0]

[2, 3, 1, 0]

Input: $V = 6$, $E = 6$, `edges[i] = [[1, 3], [2, 3], [4, 1], [4, 0], [5, 0], [5, 2]]`



Output: true

Explanation: The output true denotes that the order is valid. Few valid Topological orders for the graph are:

[4, 5, 0, 1, 2, 3]

[5, 2, 4, 0, 1, 3]